

RESEARCH SOURCES INDEX: 1517 San Juan Ave

Subject: Attic Ventilation Failure & Remediation Strategy

Usage: To support the “Belt & Suspenders” strategy and the Warranty Claim against Vintage Roofing.

1. EXPERT BUILDING SCIENCE (The “Physics” Authority)

Primary Source: Joseph Lstiburek (Building Science Corporation)

- **Video:** “To Vent or Not to Vent” by Building Science Corp
 - [Watch on YouTube](#)
 - **Key Insight:** The “Don’t Suck” Rule. If exhaust > intake, the attic depressurizes and sucks conditioned air from the house.
 - **Key Insight:** “Wash the Deck.” You must wash the entire underside of the roof deck with cold air to prevent ice dams.
- **Book:** *Builder’s Guide to Cold Climates (2022 Edition)*
 - **Key Insight:** In cold climates, balanced ventilation is safety-critical. Snow acts as an insulator (R-2/inch), requiring soffit flow to flush heat even in insulated attics.
- **Article:** “Interview with Dr. Joe Lstiburek — The Ventilation Debate Continues”
 - [Read on Energy Vanguard](#)
 - **Key Insight:** Warns against over-ventilating the living space (62.2 rates) but mandates balancing attic pressure to prevent moisture pull.

Secondary Source: “The #1 Mistake”

- **Video:** “Don’t Fall Victim to the #1 Mistake in Attic Ventilation” (https://www.youtube.com/watch?v=trgqlva_pCk)
 - **Key Insight:** Explains why upgrading to a high-flow Ridge Vent (Exhaust) without upgrading Soffits (Intake) destroys a roof.
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2. REGULATORY & LEGAL (The “Hard Law”)

Source: British Columbia Building Code (BCBC 2018/2024)

- **Link:** [BC Building Code - Section 9.19. Roof Spaces](#)
- **Section 9.19.1.2 (Ventilation Ratios):**
 - **Requirement:** 1/300 ratio mandated.

- **Distribution:** Requires 25% of vents at the TOP and 25% at the BOTTOM.
- **Section 9.19.1.3 (Clearance/Airway):**
 - **Application:** Proves that intake vents located mid-roof (SV-50s) fail to provide the mandatory airway at the eaves.
 - **CRITICAL LINK:** *This section directly defines the violation — to be documented via site observations of the soffit/eave areas.*

Source: CSA Standards (National Standard of Canada)

- **Standard:** CSA A93:19 (R2024) - Airflow ventilators for the unoccupied spaces of buildings
 - [CSA Group Store Link](#)
- **Clause 7 (Weather Infiltration):**
 - Vents are tested for rain/snow entry under specific airflow directions. Using an exhaust vent (SV-50) as an intake creates “Vacuum Infiltration” risks not covered by the certification.
- **Clause 9 (Marking):**
 - Manufacturers must mark the Net Free Area (NFA), legally forcing the roofer to do the math.

3. MANUFACTURER SPECIFICATIONS (The “Product” Data)

Source: Menzies Metal Products (SV-50 Vent) [!] KEY EVIDENCE

[!CAUTION] “Exhaust vent for mounting at upper portion of steep roof (minimum pitch 3.5:12). Not to be used as an intake vent.” — Menzies SV-50 Official Product Description

- **Product Page:** [SV50 Plastic Roof Vent](#)
- **Technical Drawing:** [PDF](#)
- **Short Specifications:** [PDF](#)
- **Installation Instructions:** [PDF](#)

Verified Specifications: | Spec | Value | |——|——| | **NFA (Net Free Area)** | 50 sq. in. per vent | | **Intended Use** | **EXHAUST ONLY** | | **Mounting Position** | Upper portion of steep roof | | **Certifications** | CSA CAN3-A93, IAPMO PS 64 |

Application to This Case: * Original roofer installed **SV-50 vents explicitly as “air intake vents”** * **From Vintage Roofing Invoice:** > “*Upgrade existing ventilation by installing 8-air intake vents using SV-50*” * Manufacturer explicitly states: “**Not to be used as an intake vent**” * **This is direct evidence of product misuse by the contractor** * Note: 12 SV-50s counted on site vs. 8 quoted — additional 4 may have been added

Source: Lomanco OmniRidge® OR-4 (The “Beast”)

- **Product Page:** [Lomanco Ridge Vents](#)
- **Model:** OR-4 (4-foot sections)
- **NFA: 18 sq. in. per linear foot** (verified across multiple sources)
- **Per 4-ft piece:** 72 sq. in. NFA

Features (from Lomanco): * External venturi baffle * Weatherproof underlapping joint * Four drain holes per foot per side * Preformed to 3/12 pitch

Calculation for This Property (FROM INVOICE): * **Invoice states:** “60’ of 13” *Omni vent*” * **Length:** 60 linear feet (CONFIRMED) * **Total Exhaust NFA:** $60 \times 18 = 1,080$ sq. in.

4. INDUSTRY CONSENSUS (The “Peer Review”)

Source: Reddit Community (r/BuildingScience)

- **Thread Topic:** “Mixing Mid-Roof Box Vents with Ridge Vents.”
 - **Consensus:** “Short Circuiting.” Using box vents (SV-50s) halfway up the roof “steals” the air from the ridge vent, leaving the soffits dead.
 - **Consensus:** “Survivor Bias.” Roofers do it because “we always have,” but they don’t see the mold that forms 2-3 years later (like in this case).
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5. PROJECT EVIDENCE (Site-Specific Data)

- **Date Stamping:** New Plywood (Installed April 2024).
 - **Significance:** Proves mold is recent (< 2 years) and directly caused by the new roof configuration, ruling out “pre-existing conditions.”
- **Intake Deficit Calculation:**
 - **Exhaust:** 1,080 Sq In NFA.
 - **Intake:** 600 Sq In NFA.
 - **Result:** 45% Deficit (Vacuum).